

Certify What You Measure: A Ground-Truth-Free Range–Null Audit for Learned Ghost and Single-Pixel Imaging

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Abstract

Ghost imaging (GI) and single-pixel imaging (SPI) reconstruct a scene from a small number of bucket measurements and have become attractive for low-light, broadband, and computational imaging. At low sampling ratios, learned reconstructors produce sharp, natural-looking images; however, it remains unclear which image content is supported by the recorded measurements and which is supplied by the prior, and this fidelity cannot be assessed when the ground truth is unknown. Here, we report a ground-truth-free measurement-consistency audit (MCA) that, from the sensing operator and the recorded bucket signals alone, certifies the measured content of any reconstruction and attributes the remainder to the prior. The audit operator is the classical Tikhonov filter-factor complement; the contribution is its ground-truth-free, per-record, reconstructor-independent use, together with a constructive feasible-but-wrong witness and a third-party forensic ledger over published pipelines. It is shown that the audit contracts the i -th measured mode of the record residual by exactly $\lambda/(\lambda + \sigma_i^2)$, independent of the reconstructor. On a snapshot spectral (CASSI) operator whose singular spectrum spans $528\times$, this contraction is demonstrated to vary over 2×10^4 (under the declared integer-shift model), the mode-resolved face of the same closed form that collapses to a single value for a flat-spectrum operator. Because most degrees of freedom are unmeasured at these sampling ratios, any record-consistent reconstruction must supply most visible detail from the null space; using the proposed audit as a forensic tool, we quantify, for the first time to our knowledge, where that detail is located by decomposing the reported gains of published deep-learning pipelines through each operator’s own projectors. The null shares are 39%, 40%, and 95% for three ghost-imaging pipelines, and 100–101% for twelve of thirteen published spectral-imaging models (exceeding 100% because the models slightly perturb the measured component), with 93% for the sole exception. The same range–null boundary is turned into a governed dial that injects prior detail only where the measurement is blind, so that $A\hat{x} = y$ is preserved exactly at every setting. It is further demonstrated that a feasible-but-wrong reconstruction can match the bucket record to a relative residual of $\sim 10^{-15}$ (unconstrained, float64), far below the $\sim 10^{-3}$ noise floor of the true scene, while depicting a different object. The proposed audit provides a ground-truth-free reliability indicator that opens up an avenue for accountable GI/SPI reconstruction at low sampling ratios.

1. Introduction

Ghost imaging and single-pixel imaging retrieve a two-dimensional scene from a sequence of structured illuminations and the corresponding single-pixel (bucket) intensities, without a spatially resolved sensor [1–6]. Since the first correlation-based demonstrations [1,2] and the computational and single-pixel variants that followed [3–6], the modality has been applied across low-light imaging, imaging through scattering media, multispectral and three-dimensional sensing, and optical security [5–9]. A recurring practical constraint is that high image quality is difficult to obtain at low sampling ratios, where the number of bucket measurements is far smaller than the number of image pixels and the reconstruction problem is deeply underdetermined.

Deep learning has become a powerful tool for this regime, and learned reconstructors routinely report large improvements in image quality over conventional correlation, back-projection, and compressed-sensing methods [10–14]. However, a learned network fills the unmeasured part of the image with content learned from a training prior. At low sampling ratios most of the visible detail is prior-supplied, and a reconstruction can be sharp, natural-looking, and fully consistent with the recorded bucket signals while depicting structure the measurement never constrained. The accuracy of such a reconstruction cannot be assessed in practical applications where the ground truth is unknown [12]; existing reliability tools estimate a statistical uncertainty from a Bayesian network [12], but do not separate, from the measurement itself, which content is measured from which is invented.

Several routes address image quality at low sampling ratios — optimized illumination bases and deep priors [10,11,13,14], compressed-sensing regularization [15,16], and range-null-space reconstruction that installs learned content in the operator’s kernel [17–19] — yet each optimizes fidelity to a ground-truth reference and none provides a ground-truth-free statement of what the bucket record actually certifies. Therefore, it is desirable to certify, without any reference image, which part of a reconstruction is supported by the measurement and which part is supplied by the prior, and to make this certificate exact and independent of the reconstructor.

In this paper, we report a ground-truth-free measurement-consistency audit (MCA) for learned ghost and single-pixel imaging, built on the range-null geometry of the sensing operator. The contributions are summarized as follows. Firstly, we show that measurement consistency cannot certify image content, and we construct a feasible-but-wrong reconstruction that matches a target bucket record to floating-point precision — more tightly than the true scene matches its own noisy record — while depicting a different object. Secondly, we develop the proposed audit, a post-hoc operator that certifies the measured content of any reconstruction with an exact per-mode contraction $\lambda/(\lambda + \sigma_i^2)$, using only the sensing operator, the recorded bucket signals, and one regularization parameter; the operator itself is the classical Tikhonov filter-factor complement, and the novelty is its ground-truth-free, per-record, reconstructor-independent use. Thirdly, the audit is used as a forensic tool to decompose the reported gains of published pipelines through each operator’s own projectors — to the best of our knowledge, the first ground-truth-free attribution of published pipelines through their own measurement operators — and it is demonstrated on a snapshot spectral operator that the per-mode contraction resolves into a 2×10^4 -fold graded profile that a flat-spectrum operator cannot exhibit. Finally, the same boundary is turned into a governed dial that supplies prior detail strictly in the unmeasured subspace, preserving the recorded bucket signals exactly. The principle is presented in Section 2, results and discussion in Section 3, and conclusions in Section 4.

2. Principle

2.1 Range–null geometry of the sensing operator

The bucket measurements of a GI/SPI system can be described by

$$y = Ax + \varepsilon,$$

where $x \in \mathbb{R}^n$ denotes the vectorized scene, $y \in \mathbb{R}^m$ denotes the bucket-measurement vector with $m \ll n$, $A \in \mathbb{R}^{m \times n}$ denotes the sensing operator whose rows are the illumination patterns, and ε denotes the measurement noise. Let $A^\dagger$ denote the Moore–Penrose pseudoinverse and let $P_R = A^\dagger A$ and $P_0 = I - A^\dagger A$ denote the orthogonal projectors onto the row space (the measured subspace) and the null space (the unmeasured subspace) of A , respectively. These projectors satisfy

$$AP_0 = 0, \quad AP_R = A,$$

so that $Ax = AP_Rx$: the recorded bucket signals are a function of the measured component $P_Rx = A^\dagger y$ alone (its noisy counterpart $A^\dagger y = P_Rx + A^\dagger \varepsilon$ under noise), while the null-space component P_0x is invisible to the sensor. Any two scenes that share the measured component therefore produce the same bucket record no matter how much they differ in the null space. In the locked GI setting used below — $n = 4096$, $m = 205$ (a 5% sampling ratio) — the measured subspace has dimension 205 and the unmeasured subspace has dimension 3891; most of what the eye reads as detail is carried by the unmeasured subspace.

2.2 Measurement consistency does not certify image content

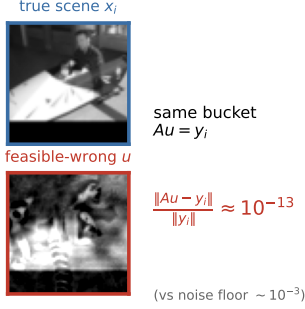
Because $AP_0 = 0$, measurement consistency is a statement about the measured subspace only. This is made concrete by a construction. For any target record y and any donor scene x_j , the feasible-but-wrong reconstruction

$$u_j(y) = x_j - A^\dagger(Ax_j - y)$$

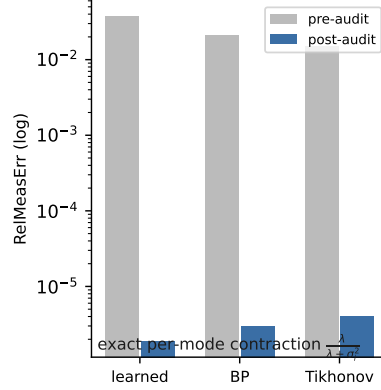
satisfies $Au_j(y) = y$ and $P_0u_j(y) = P_0x_j$: it reproduces the target bucket record exactly while carrying the donor’s null-space content. It is demonstrated that such a reconstruction matches the record to a relative residual $\|Au_j - y\|/\|y\| \approx 10^{-15}$ (unconstrained, float64) — far below the $\sim 10^{-3}$ noise floor at which the true scene matches its own noisy record — while remaining semantically far from the target (Fig. 1, left; Fig. 2). That an exact projection onto the record fiber beats a noisy-record residual is expected; the substantive point is the semantic divergence at equal-or-better consistency. Alternating the exact projection onto the record with a clip to the physical range $[0, 1]$ yields a box-legal feasible-but-wrong reconstruction whose residual remains well below the noise floor; the barrier is therefore not an artifact of unconstrained linear algebra. Consistency is not correctness, and a reliability indicator based on the record alone must certify only what the measurement constrained.

Certify what you measure, govern what you cannot

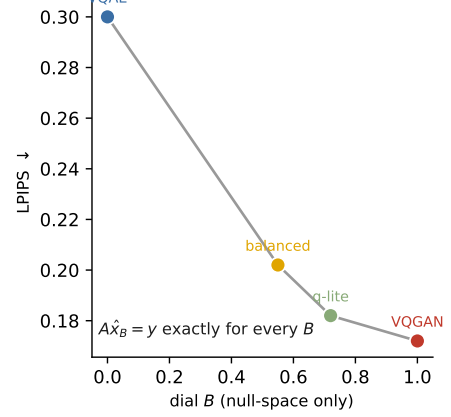
(a) Cannot certify null



(b) Can audit row record



(c) Therefore govern prior detail



consistency $\neq$ correctness

Figure 1: Fig. 1. Range-null accountability for undersampled ghost imaging. Left, a feasible-but-wrong reconstruction u that matches the target bucket record to numerical precision, so measurement consistency cannot certify the unmeasured content ($\|Au - y\|/\|y\| \approx 10^{-13}$ for the pair shown, $\sim 10^{-15}$ across pairs, versus a 10^{-3} noise floor). Middle, the proposed audit (MCA) contracts only the measured modes, with per-mode factor $\lambda/(\lambda + \sigma_i^2)$. Right, the governed dial injects prior detail only through P_0 , preserving $A\hat{x}_B = y$ while moving along the perception-distortion curve (locked: LPIPS -32.6% at -0.45 dB, 8/8 gate). GI: ghost imaging; MCA: measurement-consistency audit; u : feasible-but-wrong reconstruction; A : sensing operator; y : bucket-measurement vector; P_0 : projector onto the unmeasured (null) subspace; σ_i : singular value; λ : regularization parameter; B : dial weight; LPIPS: learned perceptual image patch similarity.

The converse in one picture: every point on the line reproduces the record;
the measurement cannot prefer the truth over the splice

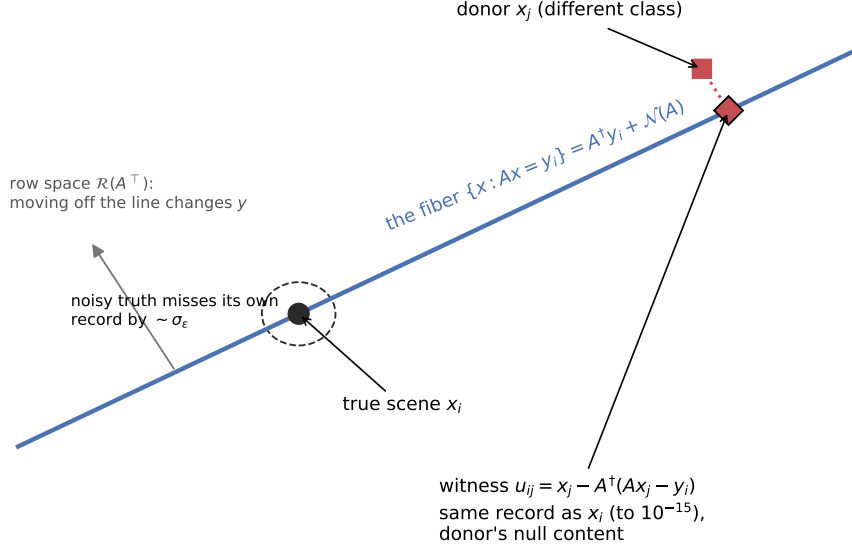


Figure 2: Fig. 2. Geometry of the feasible-but-wrong reconstruction. Every point on the line (the record fiber $\{x : Ax = y\}$) reproduces the bucket record; only motion off the line is visible to the measurement. The constructed reconstruction is the donor scene projected onto the fiber: it matches the record to $\sim 10^{-15}$, tighter than the true scene (dashed circle, radius $\sim \sigma_\epsilon$), while carrying the donor’s null-space content. A : sensing operator; y : bucket record; $\{x : Ax = y\}$: record fiber; $A^\dagger$: Moore–Penrose pseudoinverse; σ_ϵ : standard deviation of the bucket noise.

2.3 The proposed measurement-consistency audit

For any reconstruction $\hat{x}$, the proposed audit can be described by

$$\Pi_y^\lambda(\hat{x}) = \hat{x} - G_\lambda(A\hat{x} - y), \quad G_\lambda = A^\top(AA^\top + \lambda I)^{-1},$$

where $\lambda > 0$ denotes a regularization parameter, fixed at $\lambda = 1.0 \times 10^{-3}$ in all experiments below (when the noise level is known, the Morozov discrepancy principle — matching $\|A\hat{x} - y\|$ to the noise level — is the principled selection rule), and the correction $G_\lambda(A\hat{x} - y)$ lies in the row space so that $P_0 \Pi_y^\lambda(\hat{x}) = P_0 \hat{x}$: whatever a prior placed in the null space passes through untouched. The audit needs only (A, y, λ) — no ground truth and no access to the reconstructor. Diagonalizing $AA^\top$, the residual in the i -th measured singular mode of A (singular value σ_i) is scaled by

$$c_i(\lambda) = \frac{\lambda}{\lambda + \sigma_i^2},$$

which is exact, image-independent, and reconstructor-independent; as $\lambda \rightarrow 0$ the audit becomes a hard projection onto the record fiber $\{x : Ax = y\}$. The factor $c_i(\lambda)$ is the complement of the Tikhonov filter factor [20]; its use here is a post-hoc, per-record reliability indicator applied after arbitrary reconstructors. In float64 the measured contraction matches the closed form to

1.0×10^{-10} , so the audit reports, mode by mode, how far a reconstruction stands from the recorded bucket signals.

2.4 The governed null-space dial

Because the null space cannot be certified, prior detail is not hidden there but metered. For classical and generative endpoints d_A and d_G of a reconstruction, the governed dial can be obtained by

$$\hat{x}_B = A^\dagger y + P_0(d_A + B(d_G - d_A)),$$

where B denotes a scalar dial weight and the injected detail is confined to the null space, so that $A\hat{x}_B = AA^\dagger y = y$ holds exactly for every B . Algebraically, this null-space data-consistency form coincides with the range–null decomposition used by denoising diffusion null-space models [18] and by range–null-space spectral reconstruction [19]; what is new here is not the form but its use — the audit as a post-hoc, reconstructor-agnostic certificate, and the dial as a governance instrument that meters, rather than trusts, the injected content. The dial traces a perception–distortion curve while never converting the perceptual gain it produces into a measurement claim.

3. Results and discussion

3.1 The governed dial on locked ghost imaging

The proposed dial is evaluated on 64×64 grayscale STL10 scenes at a 5.0% sampling ratio ($n = 4096$, $m = 205$), using a signed, row-orthonormalized computational GI operator, with a representative generative (vector-quantized adversarial) prior for d_G , a classical linear-minimum-mean-square-error reconstruction for d_A , and a regularization parameter $\lambda = 1.0 \times 10^{-3}$. Under a pre-specified, one-shot protocol with a frozen dial weight, the balanced setting reduces the perceptual distance (LPIPS) by 32.6% relative to the projected classical endpoint at a peak-signal-to-noise-ratio (PSNR) cost of 0.45 dB, and passes all eight pre-specified acceptance conditions (a frozen gate on the LPIPS gain, the PSNR cost, the exact-consistency residual, and per-seed sign and monotonicity of the trade; 8/8 passed). The exact consistency $A\hat{x}_B = y$ is preserved to a relative measurement error of 3.6×10^{-7} (the float32 pipeline scale) at every operating point (Fig. 3). It is worth noting that the perceptual gain lives entirely on the quality axis; the exact consistency that accompanies it certifies reproduction of the record, not the correctness of what the record cannot see.

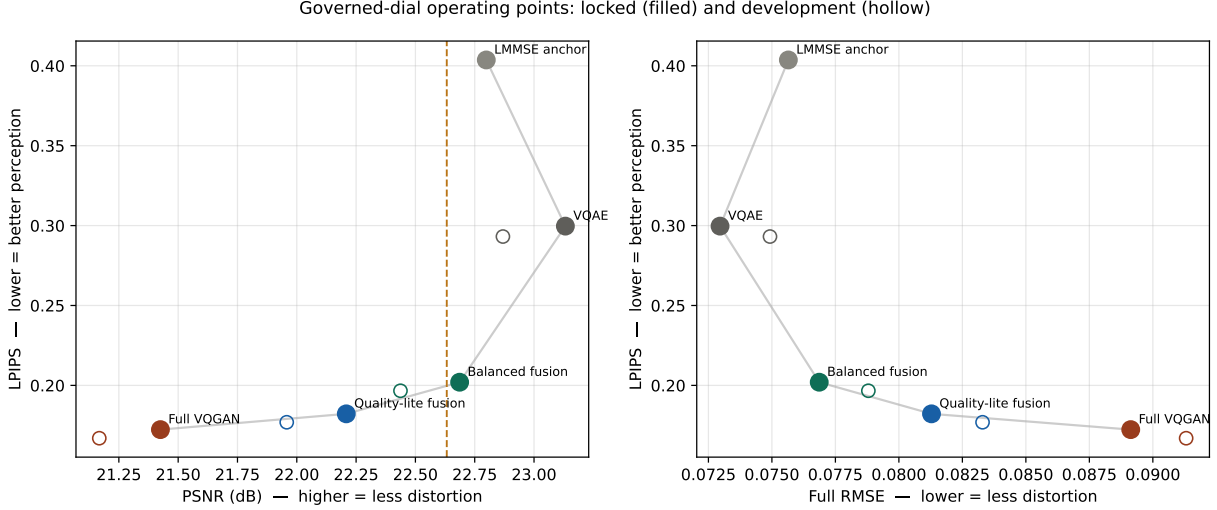


Figure 3: Fig. 3. The governed dial traces a perception–distortion curve at exact record consistency. As the dial weight B increases from 0 to 1, prior detail is injected only through the unmeasured subspace P_0 , moving the reconstruction along the LPIPS–PSNR curve while $A\hat{x}_B = y$ holds exactly (relative measurement error 3.6×10^{-7}) at every point. The balanced operating point (highlighted) reduces LPIPS by 32.6% at a PSNR cost of 0.45 dB. B : dial weight; LPIPS: learned perceptual image patch similarity; PSNR: peak signal-to-noise ratio.

3.2 Forensic decomposition of published ghost-imaging pipelines

Subsequently, the proposed audit is used as a forensic tool. At these sampling ratios most degrees of freedom are unmeasured, so any record-consistent reconstruction must supply most visible detail from the null space; the decomposition below therefore quantifies *where* each pipeline’s content is located — often close to the truth — not whether it is wrong. For three published deep-learning ghost-imaging pipelines — a pretrained-and-fine-tuned reconstructor, an untrained deep-image-prior method on a real measured speckle operator, and a self-supervised method on a genuinely noisy record — the reported improvement is decomposed, through each pipeline’s own released or measured operator, into row-space repair and null-space supply. The split is grounded in the orthogonal error-energy partition $\|\hat{x} - x\|^2 = \|P_R(\hat{x} - x)\|^2 + \|P_0(\hat{x} - x)\|^2$, a conserved quantity; the PSNR percentages reported are a monotone re-expression of it, taken relative to a range ceiling (the PSNR of the oracle row-space image $P_R x$, which requires ground truth to compute and is used only as an analysis reference, never by the audit itself). The results are shown in Fig. 4. Row-space repair saturates early, and the reported PSNR above the range ceiling is supplied from the null space; the null shares of the reported gain are 39%, 40%, and 95% (the third pipeline, a near-pure null-supplier, is an outlier), while the terminal residual error is separately 93–97% null in every case. On the real measured speckle operator, the audit drives the record residual from 3.0×10^{-3} to 3.0×10^{-15} at a PSNR change of +0.20 dB, and the measured per-mode contraction matches $\lambda/(\lambda + \sigma_i^2)$ to 3.5×10^{-11} in float64, so the closed form holds on a physically measured operator, not only on designed ones.

Projector forensics of published DL-GI pipelines, on their own operators: gains saturate the certifiable ledger and are paid from the null space

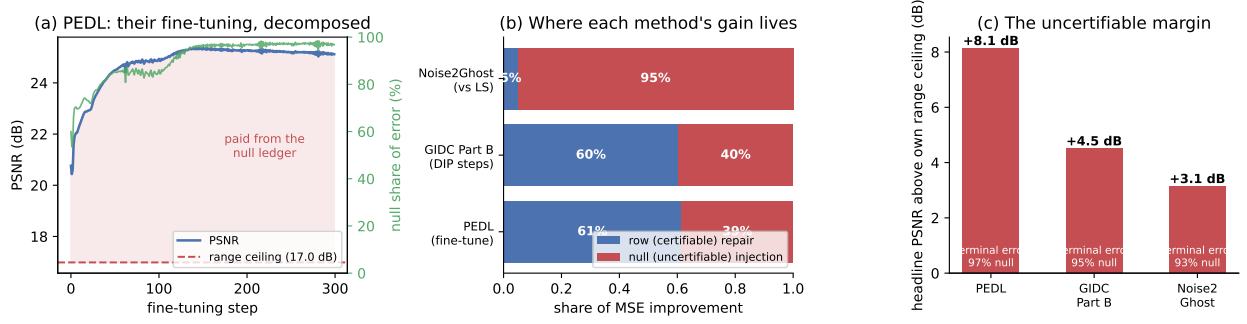


Figure 4: Fig. 4. Forensic decomposition of three published ghost-imaging pipelines on their own operators. (a) A fine-tuning trajectory decomposed per step: the PSNR rises past the operator’s range ceiling, and everything above it is supplied from the null space. (b) Attribution of each method’s improvement into row-space repair and null-space supply. (c) Every reported result stands +3.1 to +8.1 dB above its own range ceiling, with terminal error 93–97% null. GI: ghost imaging; P_{Rx} : oracle row-space (measured) image; P_0 : unmeasured subspace; PSNR: peak signal-to-noise ratio.

3.3 A non-uniform per-mode spectrum on a real coded aperture

The per-mode structure of the audit is made visible on a snapshot compressive spectral imaging (single-disperser CASSI) operator, which shifts, masks, and sums a 28-band spectral cube into a 256×310 coded snapshot — a $23.1\times$ undersampling — through the real released coded aperture (fill 0.563). Because, in the declared integer-shift model, each voxel reaches exactly one detector pixel, $AA^\top = \text{diag}(\Phi_s)$ is exactly diagonal, so the projectors and the entire singular spectrum are closed-form; a real single-disperser capture has PSF blur, cross-talk, and non-integer shifts that would break exact diagonality and smooth the spectrum, so the numbers below are operator-specific to that declared model. The measured singular values $\sigma_j = \sqrt{\Phi_s}$ over 79,360 detector modes span $[0.0093, 4.907]$ (median 3.51, a strongly left-skewed distribution), a $528\times$ ratio, and the per-mode contraction $\lambda/(\lambda + \sigma_j^2)$ at $\lambda = 10^{-3}$ therefore spans $[4.15 \times 10^{-5}, 9.21 \times 10^{-1}]$, a 2×10^4 spread (Fig. 5). This is the mode-resolved face of the same closed form that collapses to a single value for a flat-spectrum operator: for a masked-Fourier operator, as used in accelerated magnetic-resonance imaging on real data, all measured singular values equal one and the contraction is a single number 9.99×10^{-4} ; the projector identities there hold in float64 to $\sim 10^{-17}$. The two operators bracket the audit, from a flat spectrum to a strongly non-uniform one.

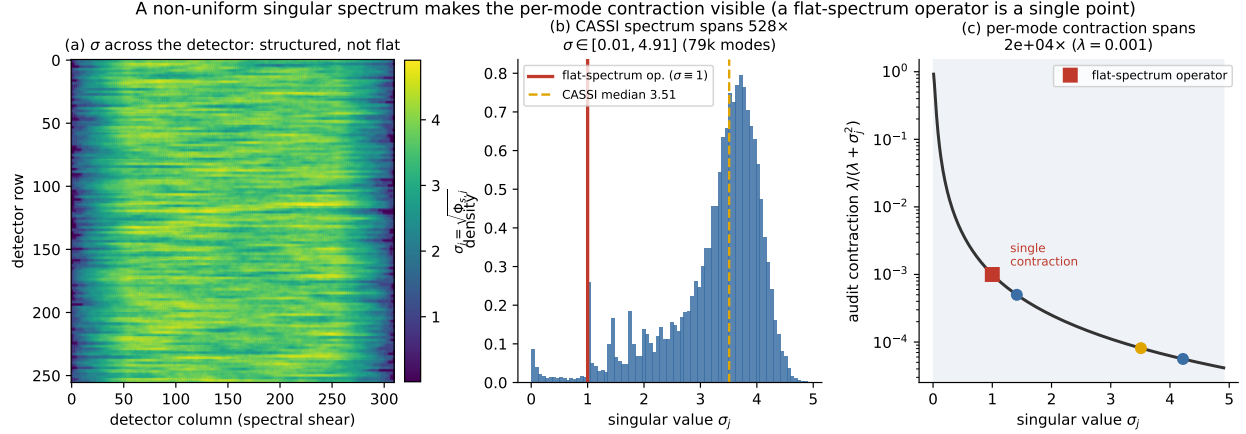


Figure 5: Fig. 5. The per-mode contraction on a real coded aperture. Left, the singular value $\sigma_j = \sqrt{\Phi_s}$ across the detector plane. Middle, the distribution of σ_j over 79,360 measured modes spans a $528\times$ range; the red line marks the single value ($\sigma = 1$) of a flat-spectrum masked-Fourier operator. Right, the per-mode audit factor $\lambda/(\lambda + \sigma_j^2)$ spreads across four orders of magnitude, where a flat-spectrum operator collapses to a single point. CASSI: coded-aperture snapshot spectral imaging; σ_j : singular value of the j -th detector mode; Φ_s : diagonal of $AA^\top$; λ : regularization parameter.

3.4 A decade of learned spectral-imaging gains in the unmeasured subspace

The forensic decomposition is applied across thirteen published spectral-imaging models on the shared coded aperture, with the wiring validated by reproducing each model’s published simulation PSNR (e.g., 34.38 versus 34.26 dB, and 38.18 versus 38.36 dB). The min-norm reconstruction $A^\dagger y$ scores 19.04 dB (the range ceiling). It is demonstrated that the models supply essentially all of their PSNR gain over the ceiling from the null space; a slightly negative row-space effect — the models mildly perturb the measured component rather than repairing it — makes the null share of twelve of the thirteen exceed 100%, at 100–101%, with GAP-Net the sole exception at 93%. The entire decade of reported progress (+6.5 dB across the field) is thus located wholly in the unmeasured subspace, and the models are 0.9–12% inconsistent with their own coded snapshot (Fig. 6). As a flat-spectrum control, the same decomposition on a masked-Fourier operator (used in accelerated magnetic-resonance imaging, where all measured singular values equal one) gives 91–92% null-located gain for an official single-coil reconstruction network, with a governed data-consistent variant retaining nearly all of that gain at exact record consistency — confirming that the ledger structure is not specific to the non-uniform CASSI spectrum. It should be emphasized that, at these sampling ratios, the null-dominance of a record-consistent estimator’s error is expected by construction and is not the finding; the finding is the contingent content that undersampling does not entail — the measured record-inconsistency of published pipelines (0.9–12%), the negative row-space effect, the specific $528\times$ non-uniform spectrum, and the agreement of the split across an entire model zoo. The ledger is an accounting of where each pipeline’s quality is manufactured, not an accusation of error; the null content these models supply is often close to the truth, but its location — entirely in the uncertifiable subspace — is what the proposed audit makes explicit.

CASSI forensics: at 23× undersampling the range ceiling is 19 dB; 100% of every model's headline gain is prior-supplied null content

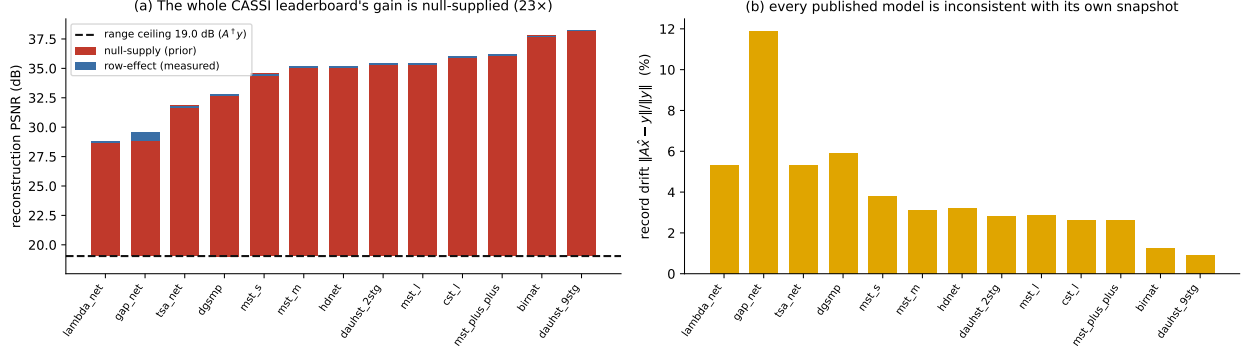


Figure 6: Fig. 6. Reported spectral-imaging gains decomposed on the shared real coded aperture. (a) Thirteen published models against the 19.04 dB min-norm range ceiling; twelve of the thirteen gains are 100–101% null-supplied (one exception at 93%), and the row-space effect is slightly negative. (b) Each model’s record inconsistency $\|A\hat{x} - y\|/\|y\|$ ranges from 0.9% to 12%. $A^\dagger y$: min-norm (range-ceiling) reconstruction; $\hat{x}$: a model reconstruction; P_0 : unmeasured subspace; PSNR: peak signal-to-noise ratio.

3.5 Scope

All operators in this work are simulation- or declared-operator-level: the GI operator is simulated; the masked-Fourier leg uses an emulated single-coil operator; and the CASSI leg uses the declared integer-shift operator on the real released aperture rather than a physical dispersive capture. The measurement noise is modeled as additive Gaussian, a first-order proxy for the signal-dependent Poisson statistics of photon counting, and no hardware validation is performed; the certificates therefore concern the declared model of the sensor. The projectors P_R, P_0 and the range-null geometry are noise-independent, so the constructive impossibility, the governed dial, and the location-of-content ledger are unchanged under any noise model. The audit’s quantitative calibration, however, is not: under signal-dependent Poisson (shot) statistics the per-mode residual variance becomes heteroscedastic, the noise floor becomes spatially varying, the discrepancy-principle λ changes, and the witness and record margins loosen from the $\sim 10^{-15}$ and $\sim 10^{-3}$ values reported here; a measured mode whose σ_i^2 is comparable to its per-mode photon-noise variance is effectively uncertified even where $c_i(\lambda)$ is small. Real GI illumination patterns are also non-negative, often differential, and DMD-quantized, whereas the locked operator here is a signed, row-orthonormalized matrix; the range-null conclusions are pattern-agnostic, while the specific spectral numbers ($528 \times, 2 \times 10^4$) are operator-specific, and Section 3.2 already shows the closed form surviving one real measured speckle operator. Shot-limited records and physically captured operators are therefore the primary next step. The unmeasured content remains uncertifiable by any within-record method — this is a property of the geometry, not an engineering gap — and the proposed audit is precisely a statement of that boundary.

4. Conclusion

We have reported a ground-truth-free measurement-consistency audit for learned ghost and single-pixel imaging, together with a constructive impossibility, a forensic decomposition of published

pipelines, and a governed null-space dial. It is demonstrated that measurement consistency cannot certify image content, that the proposed audit certifies the measured content of any reconstruction with an exact per-mode contraction $\lambda/(\lambda + \sigma_i^2)$ from the record alone, and that the reported gains of published deep-learning pipelines are largely located in the unmeasured subspace — null shares of 39%, 40%, and 95% for three ghost-imaging pipelines, and 100–101% for twelve of thirteen spectral-imaging models (with 93% for the exception). On a real coded aperture the per-mode contraction resolves into a 2×10^4 -fold graded profile that a flat-spectrum operator cannot exhibit, and the governed dial supplies prior detail only where the measurement is blind, preserving the recorded bucket signals exactly. The proposed audit provides a ground-truth-free reliability indicator that reports measurement accountability separately from image quality, and it could open up an avenue for accountable GI/SPI reconstruction in low-sampling-ratio and complex imaging conditions. Extending the audit to physically captured operators with photon-limited statistics is left to future work.

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Disclosures

The authors declare no conflicts of interest.

Data availability

The data and code underlying the results are available from the corresponding author upon reasonable request.

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